

Skills Summary

Choosing the Right Circle Theorem

Use this reference while completing your worksheet or when studying.

Skill 1 — Central and inscribed angles

Rules: a central angle equals its arc. An inscribed angle is *half* its intercepted arc, so two inscribed angles on the same arc are equal. Opposite angles of a quadrilateral inscribed in a circle are supplementary.

Tell them apart: the vertex is at the centre (central) or on the circle (inscribed).

Example: Central angle $\angle POQ$ measures 110° . Find inscribed angle $\angle PRQ$ on the same arc.

Step 1. Both angles sit on arc PQ , and the vertex of the second one is on the circle — so the inscribed-angle theorem applies and the answer is half, not equal.

Step 2. $m\angle PRQ = 110^\circ/2$

$$\Rightarrow 55^\circ$$

Try it: An inscribed angle measures 28° . Find the arc it intercepts.

check: 56

Skill 2 — Chord, secant and tangent lengths

Two chords crossing inside: the products of the two pieces are equal, $AE \cdot EB = CE \cdot ED$.

Tangent and secant from an outside point: $PT^2 = PA \cdot PB$, where PA is the *outer* part and PB the *whole* secant. Note the square — it is the step most often skipped.

Example: Two chords cross inside a circle: one is cut into 5 and x , the other into 3 and 10. Find x .

Step 1. The crossing point is *inside* the circle, so this is the intersecting-chords rule — multiply each chord's two pieces and set the products equal.

Step 2. $5x = 3 \cdot 10 = 30$

$$\Rightarrow x = 6$$

Try it: From an outside point, a tangent of length 6 and a secant with outer part 4. Find the whole secant.

check: 6

Skill 3 — Arc length and sector area

Both are fractions of a whole circle, and the fraction is the same: $\frac{\text{central angle}}{360^\circ}$.

$$\text{arc length} = \frac{\theta}{360} \cdot 2\pi r \quad \text{sector area} = \frac{\theta}{360} \cdot \pi r^2$$

Arc length carries one factor of r ; sector area carries two. That is the only difference.

Example: A sector has radius 6 and central angle 60° . Find its arc length and its area.

Step 1. The angle is a fraction of a full turn, so both answers are that same fraction of the whole circumference and of the whole area — one calculation set up twice.

Step 2. $\text{arc} = \frac{60}{360} \cdot 2\pi(6) = 2\pi$ and $\text{area} = \frac{60}{360} \cdot \pi(6)^2 = 6\pi$
 $\Rightarrow \text{arc} \approx \mathbf{6.28}$ $\text{area} \approx \mathbf{18.85}$

Try it: Find the arc length for radius 15 and central angle 48° , to the nearest hundredth.

check: 12.57

Skill 4 — The circle's equation in a configuration

Centre-radius form: $(x - h)^2 + (y - k)^2 = r^2$. If the equation shows loose x and y terms, complete both squares first — adding each completing term to *both* sides.

Once you have the centre and r , the configuration facts follow: a tangent point lies one radius from the centre, and a chord of length c sits $\sqrt{r^2 - (c/2)^2}$ from the centre.

Example: Put $x^2 + y^2 - 6x + 8y - 11 = 0$ into centre-radius form.

Step 1. The loose $-6x$ and $+8y$ terms cannot appear in centre-radius form, which is the signal to complete the square on each variable before reading anything off.

Step 2. Half of -6 is -3 and half of 8 is 4 : add 9 and 16 to both sides, giving $(x - 3)^2 + (y + 4)^2 = 11 + 9 + 16$
 $\Rightarrow (x - 3)^2 + (y + 4)^2 = \mathbf{36}$, so the centre is $(3, -4)$ and $r = 6$.

Try it: Put $x^2 + y^2 + 2x - 10y + 10 = 0$ into centre-radius form.

check: $(x + 1)^2 + (y - 5)^2 = 16$